Hypothesis Testing(1)
WMS §10.2, SJ §5.1

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Spring 2020

conflicting 2 claim → choose one → define good testing rule

Hypothesis testing

$p = 0.8$ $p < 0.8$

A claims to make 80% of his free throws. B does not believe this claim. They go to gym, A shoots 20, and makes 8.

B says: Someone who makes 80% in the long run would almost never make as few as 8 of 20, so this is evidence against A's claim.

$\boxed{\text{Idea}}$: Data (that are not compatible with a claim) are evidence against it.

Formalize:

- Model: $Y_1, \cdots, Y_{20}$ iid Bernoulli(p)
  CSS for p: (all the info about p is included in $\sum Y_i$)

$$S = \sum_{1}^{20} Y_i = \#\text{ shots made } \sim \text{ Binomial}(20, p)$$

- Hypotheses
  - $H_0 : p = 0.8$
    Null hypothesis: default state, we tend to believe it unless shown otherwise
  - $H_a : p < 0.8$
    Alternative hypothesis: research hypothesis, we do not believe it unless there is an evidence in its favor , we wish to support

<p-value>
① ask one student height.
② count number of students who are taller than one person
③ ⇒ 1% taller than the person
⇒ this person is quite tall

- Test Statistic: $S = \#$ made - random variable
- Data observed: $s = 8$ - relization
- Evidence: p-value

$$P(S \leq 8 | p = 0.8) = \sum_{s=0}^{8} \binom{20}{s} (0.8)^s (0.2)^{20-s} = 0.000102$$

assume H_0 is true.

- Conclusion:
  small p-value $\Rightarrow$ evidence against H_0
  large p-value $\Rightarrow$ no evidence against H_0

probability of result at least as extreme as result we actually got assuming H_0 is true.

p-value

- probability of a result at least as extreme (in the direction given by H_a) as the result we actually got, assuming H_0 to be true.
- p-value measures how compatible with H_0 the data are
- smaller p-values are stronger evidences against H_0 in favor of H_a

Test in practice

- Start with H_a (research hypothesis), claim that some effect or difference is present in a population

  example:

  (i) Patients live longer on average when given new drug than when given standard drug.

  (ii) Toxic substance exceeds EPA limit.

- Set up H_0 as "no effect" or "no difference" and seek evidence against H_0

  $H_0 : \mu_S = \mu_N$

  example: $H_1 : \mu_S < \mu_N$

  (i) No difference in mean survival time for patients given new and standard drug.

  (ii) No effect: level is equal to (or less than) EPA limit L.

- Test statistic measures how far data depart from what would be expected if H_0 were true.

  example:

  (i) $\bar{Y}_1 - \bar{Y}_2$ $\bar{Y}_N - \bar{Y}_S$

  (ii) $\bar{Y} - L$

in the direction of Ha

- p-value: probability of a test statistic value as extreme as what was actually observed, assuming H_0 to be true
- What is a good test statistic?
  small probability of making errors

Types of errors

Decision \ Truth	H_0	H_a
H_0	Correct	Type II error
H_a	Type 1 error	Correct

Definition

Type I error=reject H_0 when H_0 is really true
Type II error=accept H_0 when H_a is really true

$\alpha = P(\text{Type I error}) = P(\text{reject } H_0 | H_0 \text{ true})$
$\beta = P(\text{Type II error}) = P(\text{accept } H_0 | H_a \text{ true})$

Example: Legal Analogy 무죄추정의 원칙

H_0: accused innocent default state
H_a: accused guilty

Type I error=an innocent man goes to jail ← ① more serious.
Type II error=a guilty man is set free ② α가 크면 cannot get proper credit.

Rejection region = $\{(y_1, \cdots, y_n) : \bar{y} \geq 12\}$
Acceptance region = $\{(y_1, \cdots, y_n) : \bar{y} < 12\}$

Example: Y is concentration of toxic substance in river $\sim N(\mu, 5)$

Make 3 independent measurements, Y_1, Y_2, Y_3

Want to know whether $\mu > 10$ mg/dl

$$H_0 : \mu = 10 \text{ vs. } H_a : \mu > 10$$

Use rule: Reject H_0 in favor of H_a if $\bar{Y} \geq 12$ $P(\text{reject } H_0 \mid \mu = 10)$

Error probabilities:

$\alpha = P(\bar{Y} \geq 12 | H_0 \text{ is true}) = 0.0606$
$\beta = P(\bar{Y} < 12 | H_a \text{ is true})$: function of $\mu \Rightarrow \beta(11), \beta(12), \beta(15)$?

$$\alpha = P(\bar{Y} \geq 12 \mid \mu = 10) = P\left(\frac{\bar{Y}-\mu}{\sqrt{5/3}} \geq \frac{12-\mu}{\sqrt{5/3}} \,\middle|\, \mu = 10\right) = P\left(\frac{\bar{Y}-10}{\sqrt{5/3}} \geq \frac{2}{\sqrt{5/3}} \,\middle|\, \mu = 10\right)$$

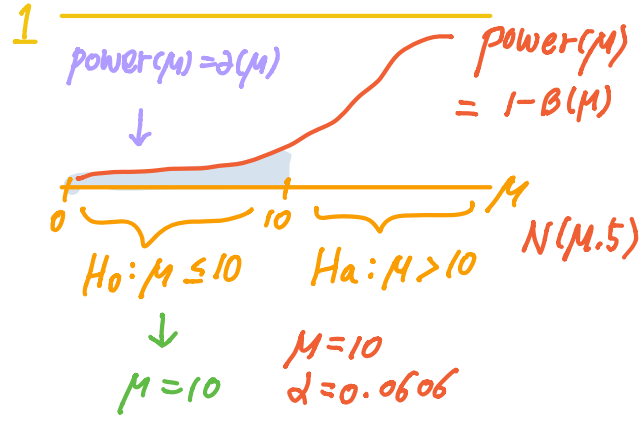
$$= P\left(Z > \frac{2}{\sqrt{5/3}}\right) = 0.0606$$

$$\beta = P(\text{do not reject } H_0 \mid \underbrace{H_a \text{ true}}_{\mu > 10})$$

depend on which μ in H_a is true.

$\beta = \beta(\mu)$

if $H_0: \mu \leq 10$ $H_a: \mu > 10$



$$\beta(11) = P(\bar{Y} < 12 \mid \mu = 11)$$

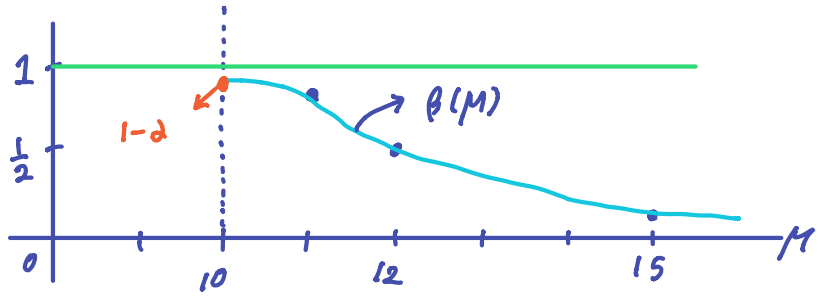
$$= P\left(\frac{\bar{Y}-11}{\sqrt{5/3}} < \frac{12-11}{\sqrt{5/3}} \,\middle|\, \mu = 11\right)$$

$$= P(Z \leq 0.77) = 0.7794$$

$$\beta(12) = P(\bar{Y} < 12 \mid \mu = 12) = \frac{1}{2}$$

$$\beta(15) = P(\bar{Y} < 12 \mid \mu = 15)$$

$$= P\left(\frac{\bar{Y}-15}{\sqrt{5/3}} < \frac{12-15}{\sqrt{5/3}} \,\middle|\, \mu = 15\right) = P(Z < -2.32) = 0.0102$$



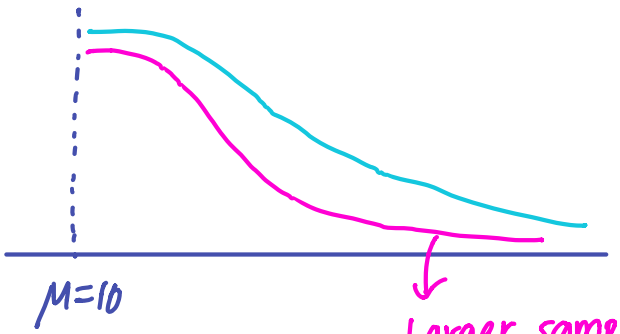
$$P(\bar{Y} < 12 \mid \mu = 10) = 1 - \alpha = 1 - 0.0606$$

$$\lim_{\mu \downarrow 10} \beta(\mu) = 1 - \alpha$$

$$\lim_{\mu \downarrow 10} \beta(\mu) = 1-\alpha$$

Max β will be $1-\alpha$, so choosing a rule with small α implies β is large for μ close to the boundary.

If we would like both small, we could take a larger sample. Larger samples reduce β for given α at any fixed μ in H_a.



Larger samples reduce β
for given α at any fixed μ in Ha.

'Good' test

we control the type I error prob by a small number (0.05, 0.01, ··)
and find a test that minimizes the type II. error probability.

$H_0 : \mu = 10$ vs $H_a : \mu > 0$

: type I error prob is a number

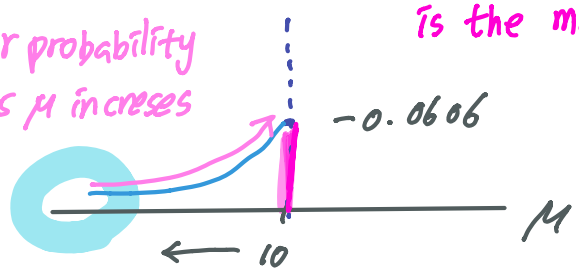
$\alpha = 0.0606$

$H_0 : \mu \le 10$ vs $H_a : \mu >$

P(reject H_0 | H_0 true)

② type I error probability at the boundary is the maximum!

① type I error probability increase as μ increses



$\mu = 0.1$

$H_a : \mu > 10$